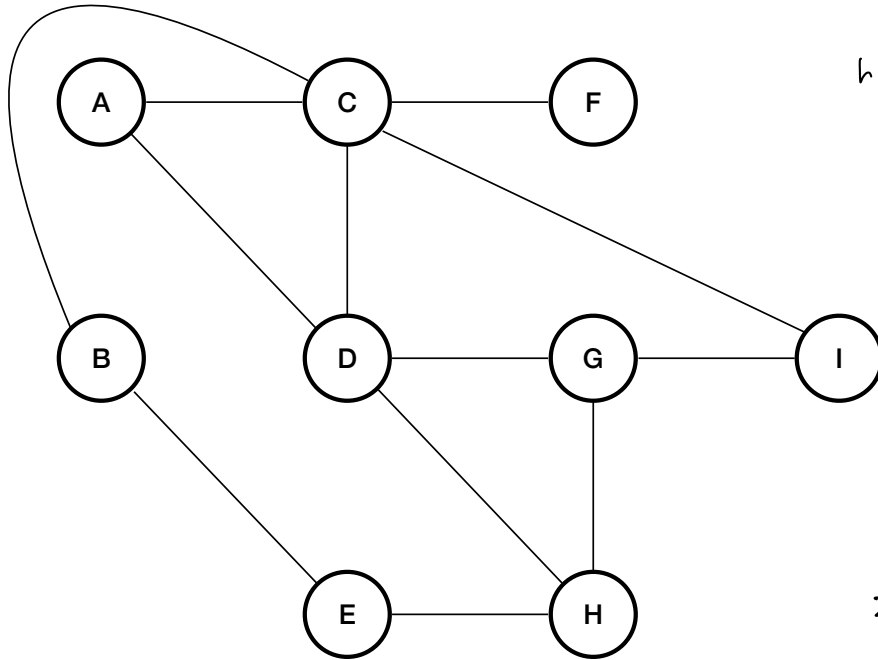
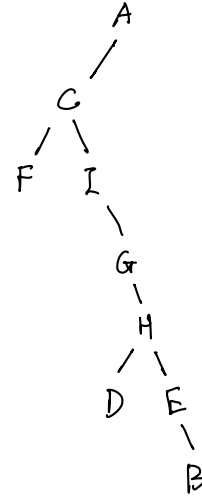


4.1 Undirected Graph

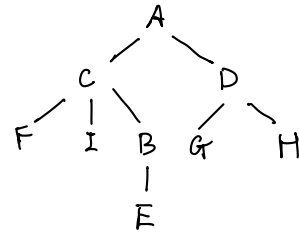


1. Run DFS on the graph starting at node A and show the exploration tree, including pre/post values. Process vertices in alphabetical order.
2. Run BFS on the graph starting at node A. Show the exploration tree.

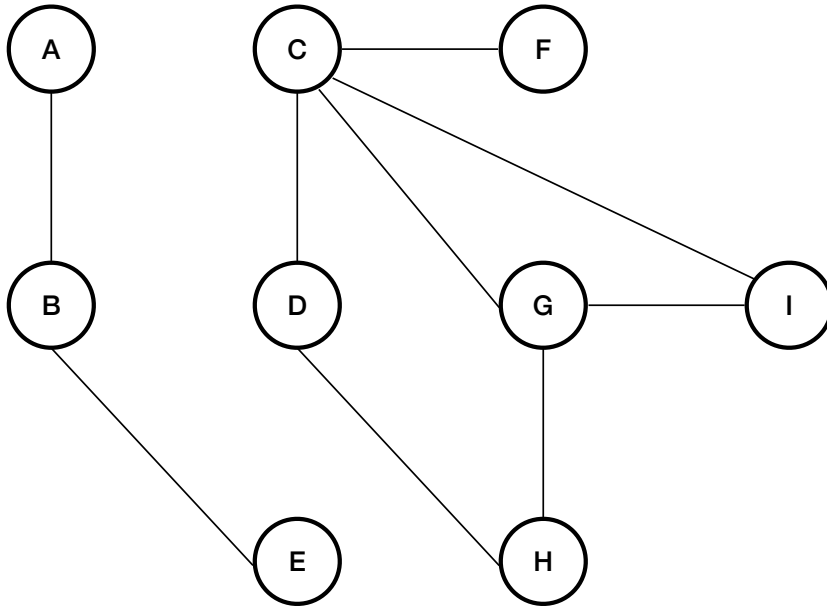
1. DFS:



2. BFS:

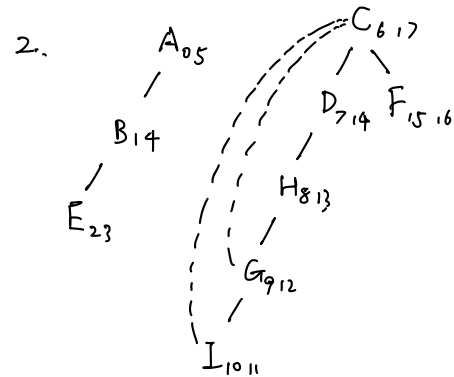
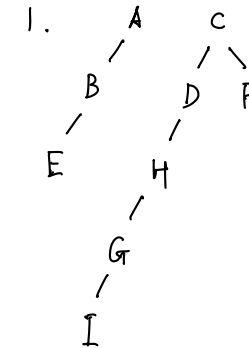


4.2 Undirected Graph

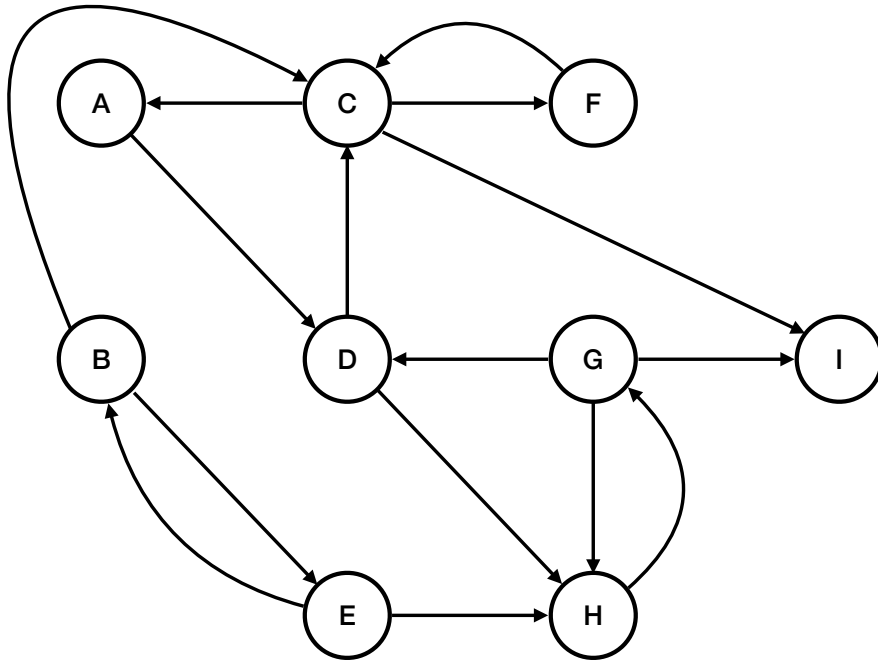


1. Run DFS on the graph starting at node A and show the exploration forest, including pre/post values. Process vertices in alphabetical order. How many connected components are in this graph?
2. Classify each edge as a tree edge or back edge and give the pre and post number of each vertex.

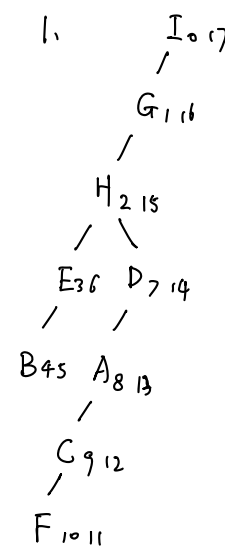
1. *2 connected components.*



4.3 Directed Graph



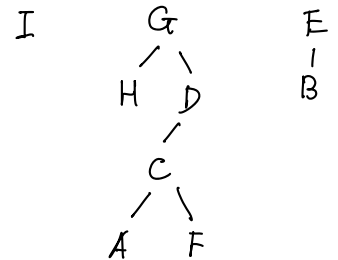
1. Starting with node I, run DFS on the reverse graph, storing pre/post
2. Run connected components on the original graph, initialize the DFS search using the vertices in decreasing order of their post numbers. Once a vertex is included in a component, it cannot be added to another. Show the connected components and the meta-graph giving the connectivity of each group. Identify which groups are sources and which sinks.



2. sorted order :

I . G . H . D . A . C . F . E . B

connected components :



meta-graph :



group **(E.B)** is a source ,
group **(I)** sinks ,